WASP-50b

R-0760

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-50_230109` · created 2026-07-28T03:54:17+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459953.606 +/- 0.082 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.158 +/- 0.046
Transit depth (Rp/Rs)^2	2.5 +/- 1.46 [%]
Orbital Inclination (inc)	82.38 +/- 2.87
Airmass coefficient 1 (a1)	1.25 +/- 0.014
Airmass coefficient 2 (a2)	-0.157 +/- 0.018
Scatter in the residuals of the lightcurve fit is	1.13 %
Best Comparison Star	None
Optimal Aperture	4.94
Optimal Annulus	17.25
Transit Duration (day)	0.025 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.158 ± 0.046 vs 0.1404 (deviation 0.38σ)
Duration fit vs published	0.025 d vs 0.0754 d (deviation 1.63σ)
Mid-time O–C	-79.9 min
Depth SNR	3.4
Residual scatter	1.13 %

Light curve

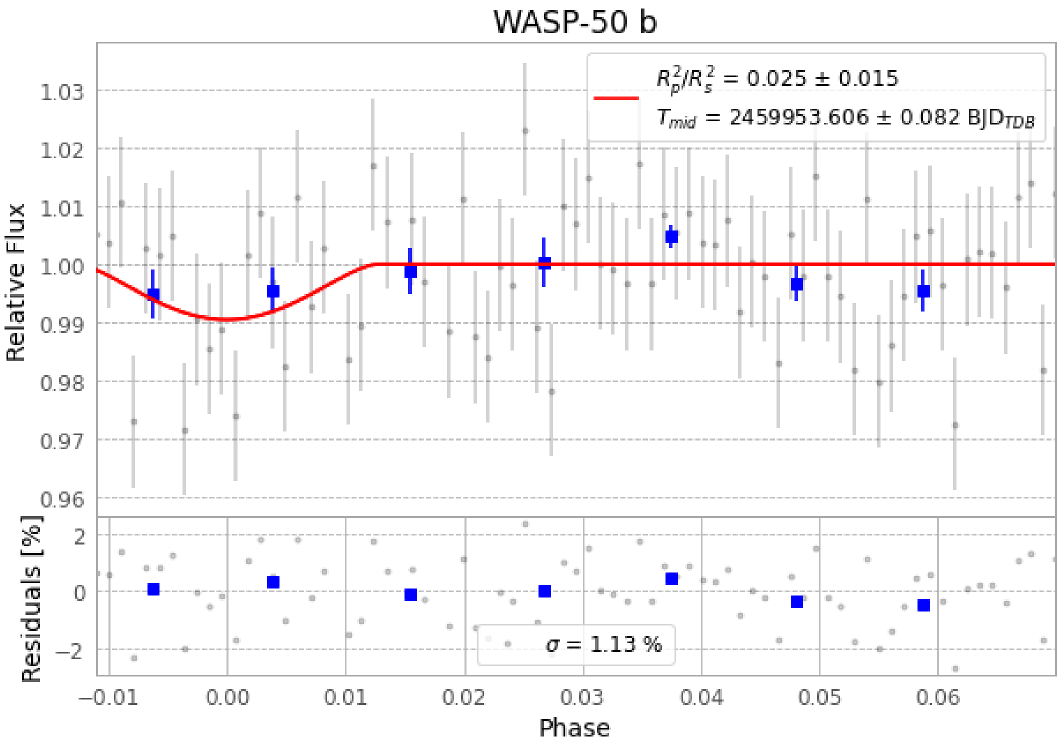

Figure 1 — FinalLightCurve_WASP-50 b_2023-01-08.png (SHA-256 54ec2218ae7102fb…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [316, 354], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 bfa4309dc1d6aeb5…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

77 FITS frames (51 MB), `WASP-50230109015715.FITS` → `WASP-50230109054515.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-230109.log (a22f927646ce…), FinalLightCurve_WASP-50 b_2023-01-08.png (54ec2218ae71…), FinalLightCurve_WASP-50 b_2023-01-08.csv (8ca7ab747656…)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-28 03:54Z.